



Maternal Stress Does Not Alter White Blood Cell Counts in Western Fence Lizard Offspring

Nathan Tsai, Emma Wen, Kate Gutierrez, David Ensminger

Biological Sciences, San José State University, San José, CA

Introduction

White blood cells are an important part in the natural immune system function and are responsible for fighting against diseases, infections, and creating antibodies. Based on the type of stress that the organism is receiving it is likely to affect the white blood cells differently as they carry out different functions. Western fence lizard white blood cells can be categorized as neutrophils, basophils, eosinophils, monocytes, and lymphocytes. Of these white blood cells, it has been shown that lymphocytes are often suppressed when exposed to chronic stress (3, 4) and neutrophils may be suppressed during acute stress (2). Therefore, in our own study we may see a decrease in lymphocytes from stressed mothers and a decrease in neutrophils from stressed offspring.



Unstained blood smear

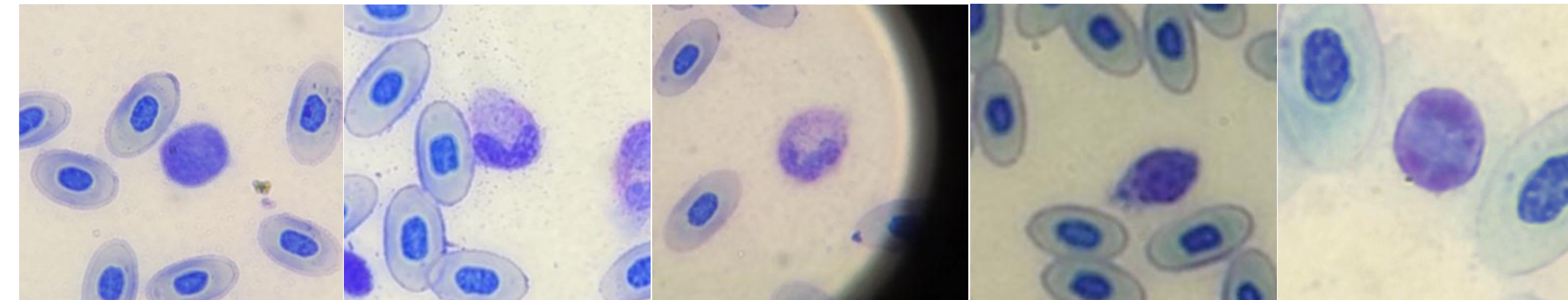
Stained blood smear

Methods

1. Gravid females captured and brought back to lab.
2. 2 treatment groups (control and CORT mother).
3. Eggs were collected from the mother and hatched.
4. Blood samples collected from hatchlings.
5. Blood smears made/stained using Wright-Giemsa.
6. Scanned slides for the first 100 white blood cells
 - a. Used a grid search pattern.
7. Statistical Analysis:
 - a. ANOVA -> variables tested are treatment, sex, and the interaction between treatment and sex. Interaction removed from model as $p > 0.3$

Results

Representative stained fence lizard white blood cells



Lymphocyte

Monocyte

Heterophil

Basophil

Eosinophil

Fig. 1

White Blood Cell Composition in Western Fence Lizard Hatchling

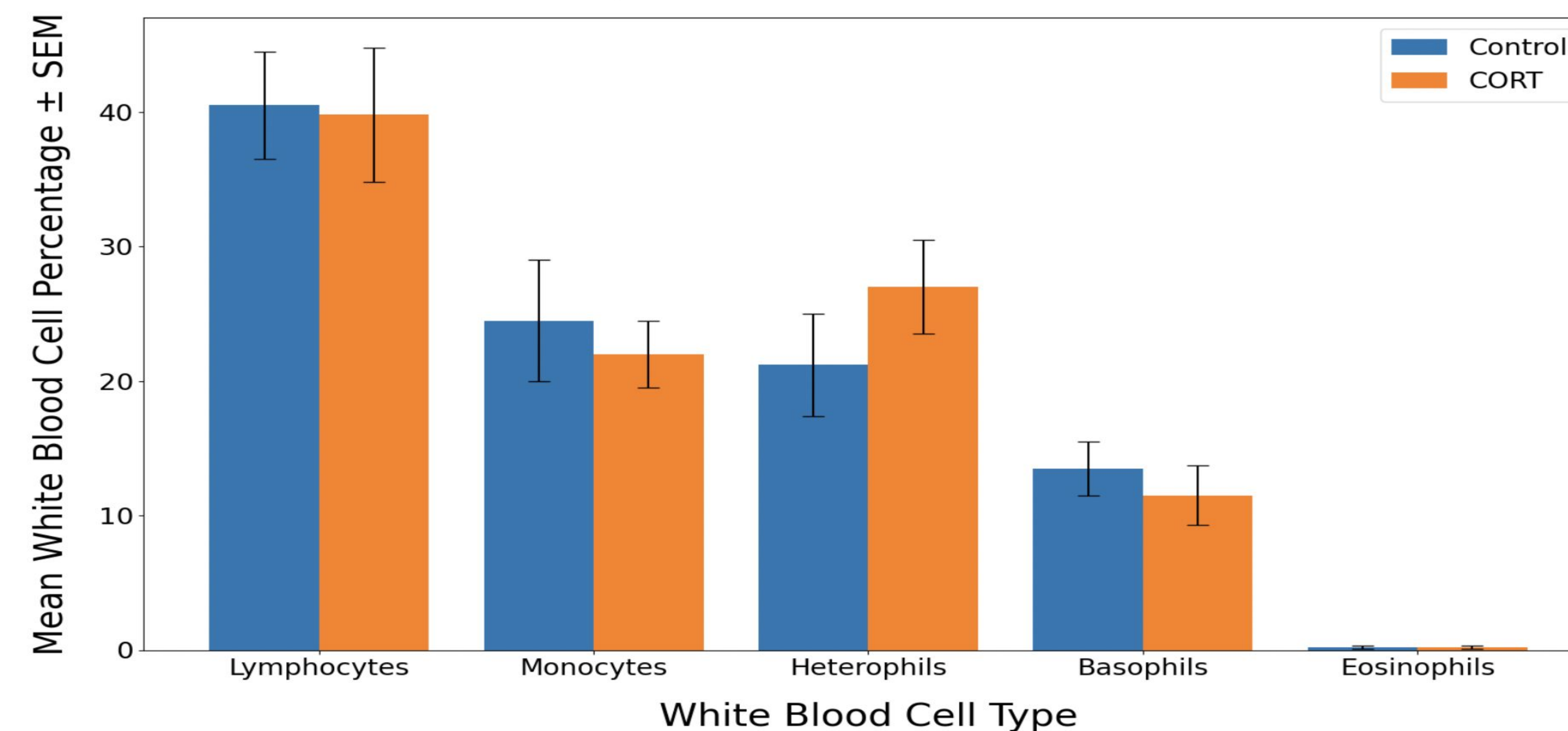


Figure 1 Legend: Relative percentages of white blood cells in offspring from control (blue bars) and corticosterone (CORT; orange bars) treated fence lizards (*Sceloporus occidentalis*). These do not sum to exactly 100%, as the bars represent the uncorrected means for the cell types within each group.

- We found no effect of maternal CORT on offspring white blood percentages (lymphocytes, $p = 0.924$, $F_{1,13} = 0.010$; monocytes, $p = 0.628$, $F_{1,13} = 0.255$; heterophils, $p = 0.282$, $F_{1,13} = 1.262$; basophils, $p = 0.530$, $F_{1,13} = 0.420$; eosinophils, $p = 0.319$, $F_{1,13} = 1.054$)

Discussion & Conclusion

While we hypothesized that maternal stress would have a negative effect on the immunity of hatchling lizards, our results did not demonstrate differences in the composition of white blood cells. This suggests that this aspect of immune function is not affected by maternal CORT. Additionally no eosinophils were found, which may suggest eosinophils represent a smaller fraction of WBC in this species or may be more fragile than other white blood cells. (4)

Future Directions

- Does this affect other aspects of immune function (ie humoral immune function or cellular immune function).
- Examine immune cell efficiency and compare to WBC count (ie BKA, LPS, KHL).



Literature Cited

1. Ensminger, D.C., et al., (2018). Journal of Animal Ecology 87, 1685–1697.
2. Ensminger, D. C., et al., (2021). Journal of Experimental Biology, 224(18)
3. Cartledge VA, Gartrell B, Jones SM.(2005) Comp Biochem Physiol A Mol Integr Physiol.141(1):100-7.
4. Tylan C, Horvat-Gordon M, Bartell PA, Langkilde T.(2020) J Exp Zool A Ecol Integr Physiol. 333(10):792-804.

Acknowledgements

- Stress Physiology Lab for animal husbandry
- San Jose State University for funding
- IDEXX BioAnalytics for assistance staining the slides